

$$\boxed{\frac{\Delta f}{\Delta x} = f'(x) + n(\Delta x)}$$

Passing to a limit in both sides of the last equation for Δx gives $\lim_{\Delta x \rightarrow 0} n(\Delta x) = 0$. Consequently, $n(\Delta x)$ is an infinitesimal (we use the same terminology as for numerical sequences, see Dialogue Three).

Conclusion: An increment Δf at a point x of a function $f(x)$ differentiable at this point can be represented by two summands, namely, a summand proportional to the increment Δx of the argument (this summand is $f'(x)\Delta x$) and a summand negligible in comparison with the first for sufficiently small Δx (this summand is $n(\Delta x)\Delta x$, where $n(\Delta x)$ is infinitesimal).

Reader: It seems that this is a formulation of the property of "linearity on a small scale" that you mentioned in the previous dialogue (see Fig. 32)

Author: Quite true. The main part of the increment of a differentiable function (a summand linear w.r.t. Δx) is called the differential of the function.